PROGRAM CODE:

```
clc;
num=poly([1],'s','coeff');
den=poly([2 1],'s','coeff');
g=syslin('c',num/den);
disp(g, 'The given first order system function G =');
t=0:0.05:50;
gs=csim('step',t,g);
subplot(121)
plot2d(t,gs)
xlabel('Time {t} (sec)')
ylabel('Step response {c(t)}')
title('Step Response of a First Order System')
gi=csim('impulse',t,g);
subplot(122)
plot2d(t,gi)
xlabel('Time {t} (sec)')
ylabel('Impulse response {c(t)}')
title('Impulse Response of a First Order System')
```

OUTPUT:

Scilab 2025.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\PRIYANKA\Documents\

Name

Documents

..
Custom Office Templates
My Music
My Pictures
My Videos
Cisco Packet Tracer.download
CS EXP 5.sci
CS EXPERIMENT 1.docx
CS EXPERIMENT 1.pdf
CS EXPERIMENT 1.sci
CS EXPERIMENT 2 .pdf
CS EXPERIMENT 2.1.sci
CS EXPERIMENT 2.2.sci
CS EXPERIMENT 3.sci
CS EXPERIMENT 4.sci
CS EXPERIMENT 5.sci
CS EXPERIMENT 6.sci
DSP EXPERIMENT 4.pdf
DSP EXPERIMENT 5.pdf
DSP EXPERIMENT 8.pdf
EXP 2.3.sci
exp 3.sci
exp 3.sci~
EXP6.1.sci
exp6.sci

File/directory filter

Scilab 2025.1.0 Console

```
  1
 ----
 2 +s
"The given first order system function G ="

-->
```

Variable Browser

Name	Value	Type	Visibility	Memory
den	1x1	Polynomial	local	264 B
g	N/A	r (Tlist)	local	1.2 kB
gi	1x1001	Double	local	8.2 kB
gs	1x1001	Double	local	8.2 kB
num	1x1	Polynomial	local	256 B
t	1x1001	Double	local	8.2 kB

Command History

```
disp(sigma,'sigma=')
_beta=atan(wd/sigma);
disp(_beta,'Beta =')
tr=(pi-_beta)/wd;
disp(tr,'Rise time tr=')
ts_2percent=4/sigma;
disp(ts_2percent,'Settling time for 2% error=')
ts_5percent=3/sigma;
disp(ts_5percent,'Settling time for 5% error=')
// -- 26/10/2026 08:18:27 -- //
// -- 26/10/2026 08:28:08 -- //
// -- 26/10/2026 08:28:15 -- //
// -- 26/10/2026 08:36:48 -- //
// -- 27/10/2026 20:07:15 -- //
// -- 30/10/2026 05:29:11 -- //
// -- 30/10/2026 05:29:15 -- //
```

News feed

Discover Scilab community

Scilab is open source.

Giving back support or use cases is appreciated.

- Add and improve functions on GitLab.
- View and edit help pages.
- Publish toolboxes on ATOMS.
- Post questions on Discourse.

27°C Partly cloudy

Search

ENG IN 05:30 30-10-2026

CS EXPERIMENT 3.sci (C:\Users\PRIYANKA\Documents\CS EXPERIMENT 3.sci) - SciNotes

File Edit Format Options Window Execute ?

CS EXPERIMENT 3.sci (C:\Users\PRIYANKA\Documents\CS EXPERIMENT 3.sci) - SciNotes

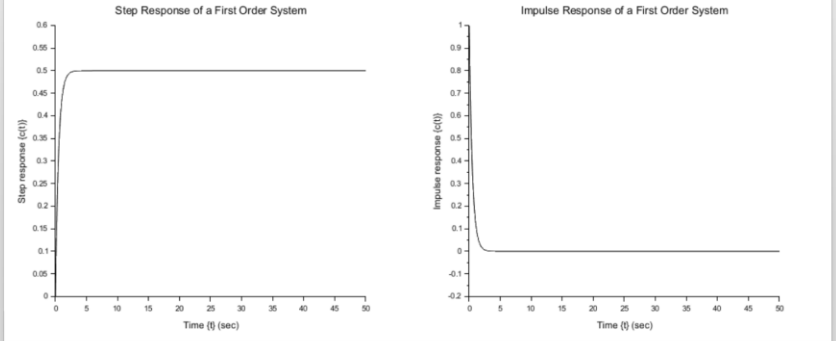
CS EXPERIMENT 3.sci | CS EXPERIMENT 2.sci | CS EXPERIMENT 3.sci

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plot2d(t,gi)
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ylabel('Impulse response (c(t))')
title('Impulse Response of a First Order System')

```

Graphic window number 0

File Tools Edit ?



24°C Mostly cloudy

Search

ENG IN 07:25 19-10-2026